VISHWAKARMA INSTITUTE OF INFORMATION TECHNOLOGY

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PROJECT REPORT

On Topic –

Seasonal Medical Storage Recommendation System and Pharmacy Process Optimisation

Submitted by -

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Abstract

Conventional management of pharmaceutical inventory is often reactive, manual and incapable of dealing effectively with the dynamic and seasonal demand for numerous medicines. This leads to excessive inventory and stock-outs, both costly. To mitigate these challenges, the authors of this paper put forward the Seasonal Medicine Recommendation & Inventory Manager (SMSRS)—an artificial intelligence (AI)-based software that changes pharmacy inventory control from reactive to proactive and data-driven. The framework employs historical sales data and combines two popular models, Meta's Prophet and SARIMA, known for their ability to capture seasonality and trend, to yield high-quality long-term demand forecasts. These models are designed to produce high quality long term demand forecasts. In particular, this system can translate detailed forecasts of demand into actionable plans. In other words, it lays out optimal order levels, schedules, and budget allocation. We made a custom data dashboard using React and TypeScript. Plus, it uses simple data visualizations. An online dashboard links to a POS system and collects feedback in real-time, making it possible to update the model after each transaction. A real-world dataset from a mid-sized pharmacy chain was used to demonstrate the system's effectiveness in automating analysis, generating consistent predictions, and enhancing forecast accuracy and inventory management efficiency. SMSRS helps pharmacies reduce operational cost significantly while improving bottom line and patient care by ensuring easy and assured availability of essential medicines.

Keywords: SMSRS, Medicine, Recommendation, Seasonal, Medical Storage, Pharmacy, Forecast, Inventory, Process optimisation.

1. Introduction

Throughout the year, the occurrence of illnesses changes and patients behave differently. This leads to more demand for drugs that can cure certain seasonal illnesses. Among the essential medicines are antipyretics, antihistamines and antivirals. The seasonal variability places undue stress on all pharmacy operations. It challenges them to design and fill their orders while achieving high service levels. Thus, this results in the drug expiry risk and excessive capital tied up in inventory. This text explains that common rules such as EOQ/ROP, Min–Max threshold, periodic review and use of safety stocks are ordinary rules assumed by most firms and proven to be unable to deal with seasonality and trend due to static nature and backward-looking problem [13-15]. Availability and waste reduction in medical supply chains that face coordination restraints and operational variability needs disciplined but adaptive control. Recent developments in pharmacy informatics and artificial intelligence suggest a more resilient operation based on data [6,13-15]. Modern evaluations recognize the value of AI in dispensing, supply planning, inventory optimization and clinical decision support systems. AI is gradually transitioning from expert systems to learning systems that can utilize vast quantities of longitudinal data [2-3]. Demand forecasting is a complex domain where classical time-series models like SARIMA and Prophet are practically strong. They can well capture seasonality, holiday effects, structural changes, and more. They offer robust baselines to the machine learning community [11-12].

To use forecasting effectively, convert the probabilistic forecasts into operational (Q, R)/(s, S) policies (i.e. optimal inventories). They consider perishability and batch sizes to reduce stockouts and expirations effectively [13-15]. Nonetheless, there is more to deploy in the real world than accuracy. Pharmacies aren't manually monitored enough due to lack of data and staff [4,6]. Standards-based integration (e.g., HL7 FHIR, RxNorm/ATC, GS1/GTIN, RFID/2D barcodes) and automation are therefore central to creating a single source of truth and to exposing timely, actionable information on item movement and availability [7,9,13–15]. At the same time, the expanding role of community and chain pharmacies toward patient-centred services increases the premium on operational efficiency:

reducing administrative load unlocks capacity for clinical activities and strengthens the patient experience [8,9]. Against this backdrop, the literature points to several unresolved needs directly aligned with this study. First, prospective validation of SARIMA/Prophet for seasonal medicines over 3–6-month horizons remains limited in pharmacy contexts, particularly where forecasts are tightly coupled to perishability- and service-level–aware control policies with end-to-end performance reporting [11–15]. Second, evidence for real-time, POS-integrated planning systems that concurrently update forecasts and inventory policies and translate them into operational (Q, R)/ (s, S) actions at scale is sparse outside single-site case studies [3,6,7,9,13–15].

Third, few deployments demonstrate fully automated, low-touch pipelines from ingestion and feature engineering to visualisation with explainable recommendations that measurably reduce manual effort and decision latency for pharmacy users [2,7,9]. Fourth, robust evaluations isolating the incremental economic and service impact of forecast-driven control over traditional Min–Max/EOQ baselines are limited in pharmacy settings [6,11–15]. Finally, there is a gap in validated approaches for data-quality robustness in live POS streams that handle zero inflation, backorders vs. true demand, outliers, and changepoints, and quantify their effect on seasonal demand forecasts and reorder decisions [4,6,11–13]. This paper presents the Seasonal Medical Storage Recommendation System and Pharmacy Process Optimisation, an end-to-end platform that operationalises state-of-the-art forecasting and inventory control for seasonal medicines. The project pursues four objectives: (1) build a forecasting engine using SARIMA and Prophet to predict seasonal medicine demand over 3–6 months; (2) integrate forecasts with real-time POS data to update inventory plans concurrently and optimize replenishment decisions; (3) create a fully automated pipeline from data ingestion to visualization with minimal manual effort and an easy-to-understand dashboard that supports data-driven decision making; and (4) demonstrate that accurate forecasting reduces costs and stockout risk, thereby ensuring medicine availability and strengthening patient confidence. Contributions to literature are seasonal forecasting and service level constrained control policies that essentially contribute to a standards-ready data pipeline for polices effectiveness on inflations. From a calibration perspective, we calibrate seasonally forecasts (SARIMA/Prophet) with service level constrained control policies (Q, R)/(s, S) that the practitioners are familiar with for implementation both implicitly and explicitly considering lead time and perishability for reorder points and order quantities [11-15].

From a standards-ready implementation perspective, we implement a standards-ready data pipeline that accepts sales data and boasts a rigorous preprocessing endeavor for zero-inflated and shock-affected time series data. It features an automatically determined methodology for subsequent feature engineering, model fitting, prediction generation, and policy calculation. It even recommends via an explainable dashboard [7, 9, 13-15]. Operationally, we evaluate outcomes that matter to pharmacies' stockout/fill rates, expiry and wastage costs, inventory turns and working capital alongside forecast accuracy using point and probabilistic metrics appropriate for seasonal horizons [11–15]. Strategically, we position the system within ongoing efforts to modernise pharmacy management systems through automation and analytics, reducing manual workload and variability while supporting the sector’s shift toward patient-centred services [6–9,13–15]. By directly addressing documented gaps, prospective validation of seasonal forecasts linked to policy, real-time POS-integrated planning, fully automated and usable pipelines, rigorous economic and service evaluations, and data-quality robustness in production, the study aims to advance both scholarship and practice. The expected impact is twofold: improved operational performance through fewer stockouts and expiries with lower working capital, and enhanced patient trust through reliable availability of seasonal medicines. In doing so, this work offers an evidence-based template for AI-enabled pharmacy inventory optimisation that is compatible with existing governance (e.g., ABC–VEN) and control frameworks, yet upgrades them with forecast-driven, real-time, and explainable decision support [1,7,13–15].

1.1 Objectives

- We'll build a forecasting engine using SARIMA and Prophet models that can be used to forecast the demand of seasonal medicine for the next 3-6 months.
- Plan with All Recommendations on the Forecast Used for Plan Integration and POS Data - Link to Make Inventory Updates Concurrently, in Real-Time and Optimise Forecasts.
- Create a pipeline from data ingestion to visualisation in a completely automated manner with minimum manual effort and an easy-to-understand dashboard for all kinds of data-driven decision making.
- Through accurate forecasting, you can reduce costs and the chances of stockout. This will ensure the availability of medicines and improve patient confidence.

2. Literature Review

The foundation of this project is built upon research in healthcare inventory management, pharmacy informatics, artificial intelligence, and demand forecasting [1–3,6,11–15]. The following synthesis situates the Seasonal Medicine Recommender & Inventory Manager academically and practically.

Healthcare inventory management: sustainability, control, and classic policies. Sustainability and efficiency are paramount in healthcare inventory management. Krishna's seven-dimensional Inventory Control Review Model identifies internal risk points: Expiration, Wastage, Redundancies, Unmanned Inventory, Control Deficiencies, Capacity Mismanagement, and KSA of manpower and provides a structured approach for auditing and improving stock stewardship to reduce costs and environmental impact [1]. In practice, pharmacy inventory control still frequently relies on classic methods: ABC/FSN/VEN prioritisation, EOQ/ROP economics, Min–Max thresholds, and periodic review [13–15]. Soraya et al. demonstrate at the hospital level that combining ABC–VEN classification with EOQ and ROP yields measurable efficiency and cost benefits in drug inventory, even with conventional methods [15]. Pfohl's contemporary chapter on Inventory Management further summarises control policies (Q, R), periodic review (s, S), and base stock, order cycle trade-offs, and service level management in a healthcare context [13]. Sharma details the managerial need for formal inventory control systems that underpin these policies [14]. Khokhar highlights the operational challenges particular to medical supply chains that make disciplined control essential [6].

Our system is designed to automate the "control deficiency" and "capacity mismanagement" dimensions by replacing static thresholds with forecast-driven, service-level-constrained replenishment proposals, while preserving compatibility with familiar control policies (e.g., (Q, R) or (s, S)) and ABC–VEN governance [1,13–15].

Forecasting demand for medicines: methods, evaluation, and policy linkage. Medicine demand displays a spectrum of patterns: intermittent/low volume SKUs, promotional pulses, seasonal pathologies, and structural changes (e.g., formulary switches). Literature supports matching the method to the signal:

- Intermittent or sparse demand: Croston and bias-corrected variants (SBA/TSB) reduce the tendency of exponential smoothing to under-forecast zero-inflated series [13,14].
- Seasonal/trended demand: classical time series models such as SARIMA, TBATS, and Prophet explicitly model seasonality and holiday/campaign effects. Dubey et al. report that SARIMA can capture seasonal structure in energy time series and provide robust baselines alongside deep learning (LSTM) [11]. Žunić et al. show Facebook Prophet's practicality for production-

grade forecasting, highlighting automatic seasonality/holiday effects and change point detection that are useful when calendars (e.g., flu season, Ramadan, public holidays) affect pharmacy sales [12].

- Machine learning/deep learning: tree-based models (XGBoost/LightGBM) and neural models (LSTM, N-BEATS) can outperform classical approaches where relationships are non-linear and exogenous variables (e.g., temperature, RSV prevalence) are available [11,12].

For service level targeting and perishables, probabilistic forecasting is preferred so replenishment can be set to the desired fill rate while controlling expiries; evaluation should therefore combine point metrics (MAE, RMSE, sMAPE/MASE) and probabilistic metrics (CRPS/WIS, interval coverage) and be verified against hold-out horizons that align with lead times and shelf life windows [11–13,15].

Crucially, forecasts must drive inventory policy. Literature recommends coupling probabilistic forecasts to service-level-constrained policies-based stock or (s, S)/ (Q, R) and adapting EOQ/ROP to include perishability, batch sizes, supplier calendars, and multi-echelon constraints [13–15]. Our system implements this link end to end: it produces calibrated forecasts (leveraging SARIMA/Prophet as baselines) and converts them into replenishment proposals aligned to target service levels and expiry risk [11–15].

AI in pharmacy: from expert systems to learning systems. AI in pharmacy has evolved from early expert systems knowledge bases with inference engines for diagnosis and drug therapy monitoring to today's data-driven learning systems that manage complex operational and clinical data at scale [2,3]. Modern reviews emphasise AI's roles across dispensing, inventory optimisation, supply planning, and clinical decision support, with documented gains in efficiency and safety [2]. Beyond classical ML, knowledge graph approaches, and graph neural networks are being actively explored for longitudinal clinical decision support; for example, KGDNet combines GNNs with attention over EHR sequences to recommend safe medication regimens while controlling drug–drug interactions [5]. While our aim is inventory rather than patient-level prescribing, the core principle of AI-driven, context-aware recommendations learned from longitudinal data transfers directly to procurement optimisation [2,5,7].

Upgrading pharmacy systems: vision, standards, and automation. Legacy pharmacy systems are widely acknowledged as insufficient for modern data-driven operations; calls to adopt "systems for the future" that integrate clinical and operational data date back decades and are echoed by recent PMS frameworks that stress automation, analytics, and safety [3,7,9]. In medical supply chains specifically, researchers report a positive relationship between real-time inventory automation and supply chain efficiency, and advocate systems that expose accurate, timely information on item location, availability and movement [6,7,13–15]. Our architecture operationalises this guidance via automated pipelines from sales ingest and feature engineering, through forecasting and policy computation, to replenishment dashboards and alerts, reducing manual work and process variability [7,13–15].

Limits of static Min–Max and advantages of forecast-driven control. Traditional Min–Max rules perform poorly under seasonality, trend, and intermittency, because thresholds are fixed and backwards-looking [13,14]. Literature and practitioner experience converge on combining probabilistic forecasts (SARIMA/TBATS/Prophet for seasonal items; Croston/TSB for sparse items) with service level–constrained (s, S) or base stock policies to reduce both stockouts and expiries, an effect particularly pronounced for short-dated medicines [11–15]. Our system embodies that shift: static thresholds are replaced by dynamically updated reorder points/quantiles conditioned on forecast uncertainty, lead time, and shelf life [11–15].

Supply chain realities: data quality, governance, and bullwhip effect. Forecasting in healthcare is routinely impeded by financial constraints, data problems, and staffing shortages, leading to delayed procurement and stockouts. Qualitative work in LMICs reinforces that poor data quality (missingness, untimely reporting, false positives), weak coordination, and insufficient IT capacity are root causes of sub-optimal forecasting and replenishment [4,6]. To be effective, systems must therefore handle:

- Data quality: zero inflation, backorders vs true demand, outliers, changepoints (e.g., pandemics), seasonality/promo effects, and denominator shifts (e.g., new branch opening). Our pipeline includes robust preprocessing (outlier capping, intermittent series identification) and drift monitoring [11–13].
- Interoperability and standards: HL7 FHIR for transactional interoperability; RxNorm/ATC/SNOMED CT for coding and therapeutic equivalence; GS1/GTIN, RFID/2D barcodes, and FMD/serialisation for tracking. The system is designed to integrate with these to ensure a "single source of truth" master data [7,9,13–15].
- Privacy/ethics and governance: GDPR/HIPAA compliance; role-based access, audit trails, explainability of recommendations (e.g., quantile contributions, demand drivers), and fairness (avoid disadvantaging low-volume essential SKUs) [2,9].

Operational efficiency and the pharmacy's expanding role: Strategic and business context. Increasingly, community and chain pharmacies are changing from being pure dispensers to becoming patient-centred, expanding their focus on prevention and chronic disease support. To make this shift, the administrative load must be automated to free pharmacists' time. Research shows that efficiency can unlock new services, improve conditions in the workplace, and so on [8,9]. Our system relieves the burden of inventory management while enhancing finances (by holding less capital in stock, reducing expiring/waste, and improving turns) [6,8,13–15].

Evidence of economic benefit and a baseline for improvement. Even when applying traditional quantitative methods (EOQ/ROP, continuous review), hospital pharmacy case studies show material savings and service improvements; for example, ABC–VEN+EOQ interventions have demonstrated expiry and cost reductions in hospital settings [15]. This establishes a baseline from which AI-enabled, forecast-driven optimisation can reasonably be expected to deliver further benefits. Our system strengthens this trajectory by adding seasonality/intermittency-aware forecasting (SARIMA/Prophet), probabilistic sizing to target service levels, and expiry-aware ordering [11–15].

Frameworks that call for AI in inventory optimisation. Recent frameworks argue that predictive analytics and ML can transform pharmacy inventory by reducing waste, improving financials, and enhancing patient outcomes [2,6–9,13–15]. The present project operationalises these frameworks into a working tool: data pipelines, fit-for-purpose forecasting models, policy optimisation, and decision dashboards [7,13–15].

Human factors and adoption (TPB). Successful adoption hinges on human factors. The Theory of Planned Behaviour suggests intentions depend on attitudes, subjective norms, and perceived behavioural control [10]. We address these by providing clear visualisations (e.g., forecast uncertainty bands and service level impacts), explainable recommendations (e.g., reorder rationale tied to quantiles and lead times), and actionable dashboards (e.g., "next best order" with expiry risk). These design choices improve attitudes toward data-driven decisions and perceived control, increasing the likelihood of sustained use [10].

Summary of measurable outcomes. Consistent with the literature, we will evaluate: stockout rate/fill rate (service level), expired units and wastage cost, inventory turns and working capital, forecast accuracy (MAE/RMSE/sMAPE/MASE, CRPS/coverage), and process KPIs (time to order recommendation, exception rate) [11–15].

2.1 Research Gap

- Prospective validation of SARIMA/Prophet for seasonal medicines over 3–6-month horizons: Limited pharmacy-domain studies link these forecasts to service-level/perishability-aware policies and report end-to-end performance in practice [11,12,13–15].
- Real-time, POS-integrated planning: Evidence is sparse for systems that concurrently update forecasts and inventory plans from POS streams and translate them into operational (Q, R)/ (s, S) actions at scale outside single-hospital case studies [6,7,13–15,3,9].
- Fully automated pipeline and usable dashboards: Few deployments demonstrate a low-touch pipeline from ingestion to visualisation with explainable recommendations that measurably reduce manual effort and decision latency for pharmacy users [7,9,2].
- Economic and service outcomes vs. Min–Max baselines: Robust evaluations isolating the incremental impact of forecast-driven control on stockouts, expiries, inventory turns, and costs beyond traditional EOQ/ROP or Min–Max are limited in pharmacy settings [13–15,6,11,12].
- Data-quality robustness in live POS data: There is a gap in validated approaches that handle zero inflation, backorders vs true demand, outliers, and changepoints in production, and quantify their effect on forecast accuracy and reorder decisions for seasonal SKUs [4,6,11–13].

3. Proposed System

To address the identified gaps, we propose the Seasonal Medicine Recommender & Inventory Manager (SMSRS), a full-stack, AI-powered software solution that transforms pharmacy inventory control from reactive to preventive, data-driven planning. The subsequent section details system architecture, system operation, implementation and algorithms.

3.1 System Architecture

SMSRS is based on a state-of-the-art three-tier architecture for better future scalability and maintainability of the system: Frontend Layer, Backend API Layer and Data Layer. Each component is deployable independently, and it is assumed that they communicate through a well-defined RESTful API.

- Frontend Layer: A React + TypeScript single-page application (SPA) renders dashboards, charts, and recommendations, and initiates data uploads and forecast requests.
- The Backend API Layer consists of a Flask service that applies business rules for data ingestion/validation, seasonality analysis, and dual forecasting (SARIMA and Prophet) to derive recommendations. It provides authenticated endpoints for file uploading, forecasting, and report access.
- Data Layer: A normalised SQLite database stores cleaned sales records, model metadata, and recommendation outputs. CSV ingestion is supported with a preprocessing pipeline.

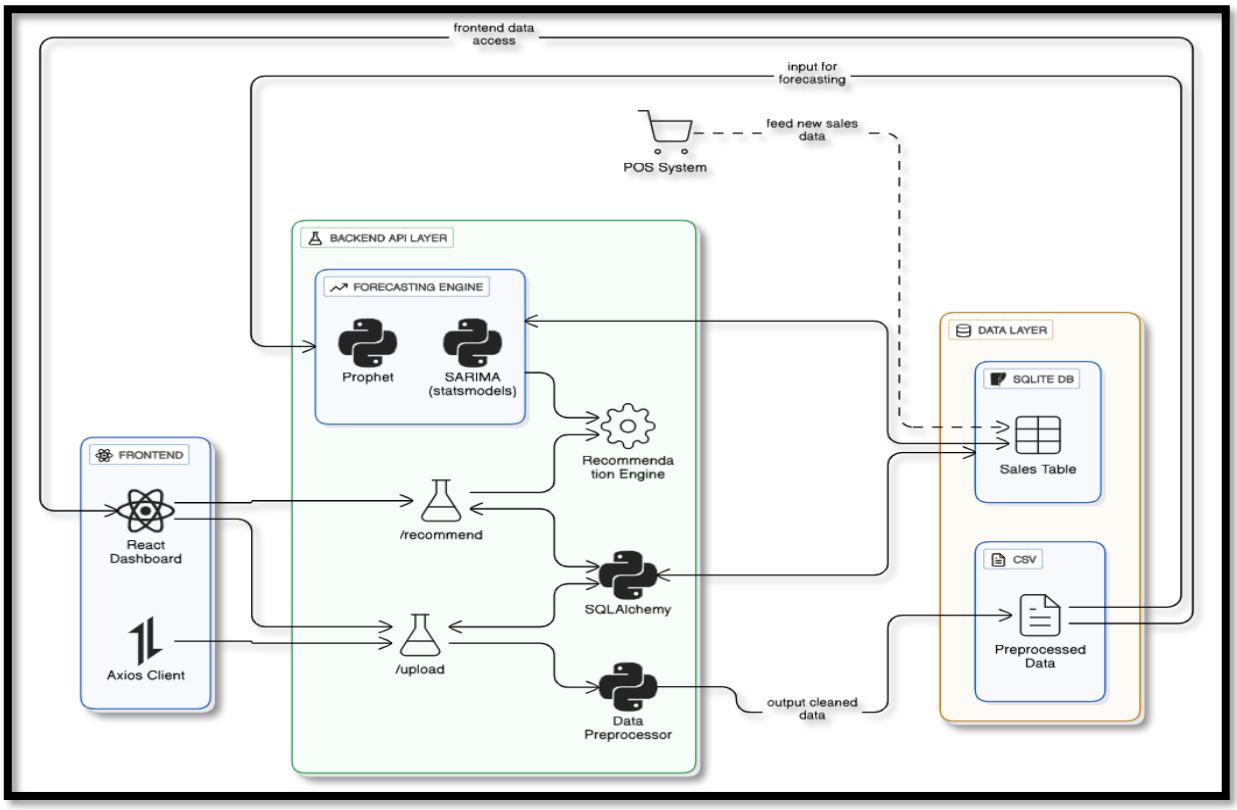


Figure 1: *System Architecture – a view of the system end-to-end (frontend, backend, and data) with major components and data movements.*

The Data Analysis and Forecasting Module serves as the backend's "brains". It accepts raw CSV data of sales, validates and cleans it, analyses seasonality, and ultimately produces predictions via SARIMA and Prophet by preparing and inputting the cleaned data. A Recommendation Engine collates the predictions into procurement plans. The API and Frontend Communication Module ensures secure transmission between the interface and backend.

3.2 Working and Implementation

Users upload CSV files with sales (medicine ID/name, quantity, unit price, invoice date, etc.) using the dashboard.

Data Ingestion and Validation

- Users can upload CSV files containing sales data (medicine id/name, quantity, unit price, invoice/date, etc.) from the main dashboard.
- The backend is validated through a validation schema to ensure date type validation, no missing entries, no duplicates and no outliers; otherwise, it provides appropriate debugging responses.
- Cleaned data persisted to SQLite and versioned for traceability.

Seasonal Analysis

- Sales are aggregated at configurable granularities (daily → weekly for SARIMA stability; daily for Prophet).
- The system computes seasonal summaries by Indian climatic periods: Summer (February–May), Monsoon (June–September), Winter (October–January), alongside rolling trends and fast-mover flags.

Forecasting Engine

For each medicine identified as a high-priority and seasonal:

- SARIMA (statistical models): Automatic grid search over plausible $(p, d, q) \times (P, D, Q)_s$It is guided by the Akaike Information Criterion (AIC) to select the best fit per series. Data are resampled to weekly frequency, requiring at least 24 weeks for reliable estimation; 12-week forecasts with 95% CIs are produced via maximum likelihood.
- Prophet: Data are formatted as (ds, y). Prophet decomposes the series into trend, seasonality (weekly, yearly), and holiday effects, with Bayesian changepoint detection. Daily runs require at least 30 days and produce 3-month-ahead forecasts with uncertainty intervals.

Assembling and validation:

- The system computes Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) via rolling-window time-series cross-validation.
- Final predictions are either confidence-weighted or simple averages of SARIMA and Prophet, with confidence scores surfaced to the UI.

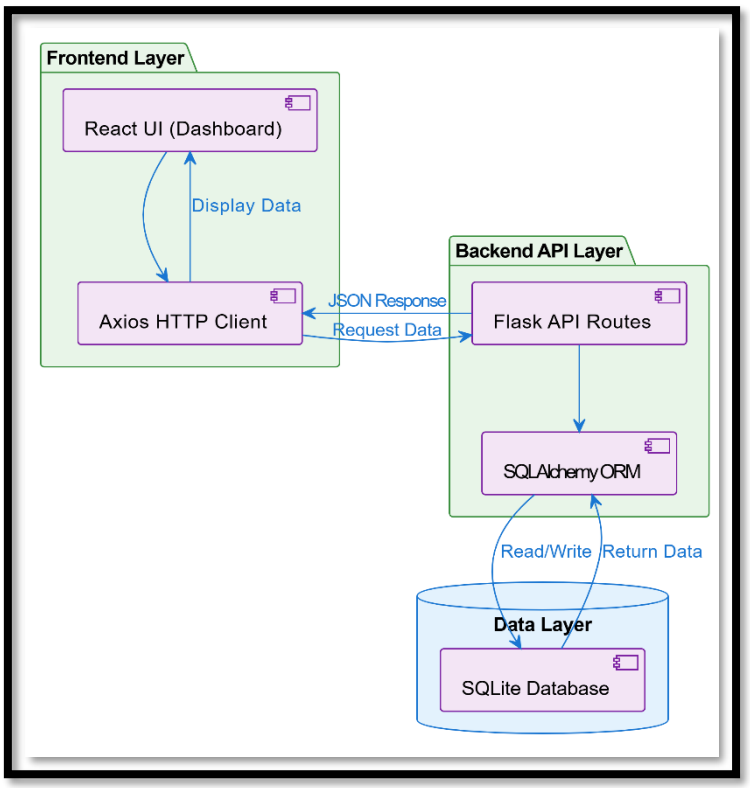


Figure 2: *Application Architecture - request/response interactions between React client, Flask API, SQLAlchemy ORM, and SQLite.*

Recommendation Generation

Numerical forecasts are converted to business actions:

- Dynamic safety stock derived from forecast variance and target service level.

- Optimal order quantity accounts for forecasted demand, safety stock, and on-hand inventory; ordering timelines reflect lead time constraints; budget estimates multiply recommended quantities by unit prices.
- Priority tags (Critical/High/Medium/Low) help managers sequence procurement.

Frontend Visualisation and User Interaction

- React dashboard (Recharts visualisations) presents seasonal decompositions, confidence intervals, stock cover days, and recommended order schedules.
- Users can filter by medicine, category, time window; export recommendations as PDF/Excel/JSON; and trigger reforecasts as new POS transactions arrive.
- All data are obtained through secure RESTful endpoints (e.g., POST /upload, POST /recommend, GET /recommend/id, GET /metrics).

Security, Reliability, and Operations

- Input sanitisation, file-type/size checks, and API authentication safeguard the pipeline.
- Idempotent ingestion and transactional writes prevent partial updates.
- Structured logging and error codes support debuggability; background jobs can be scheduled for periodic re-training.

Use Cases

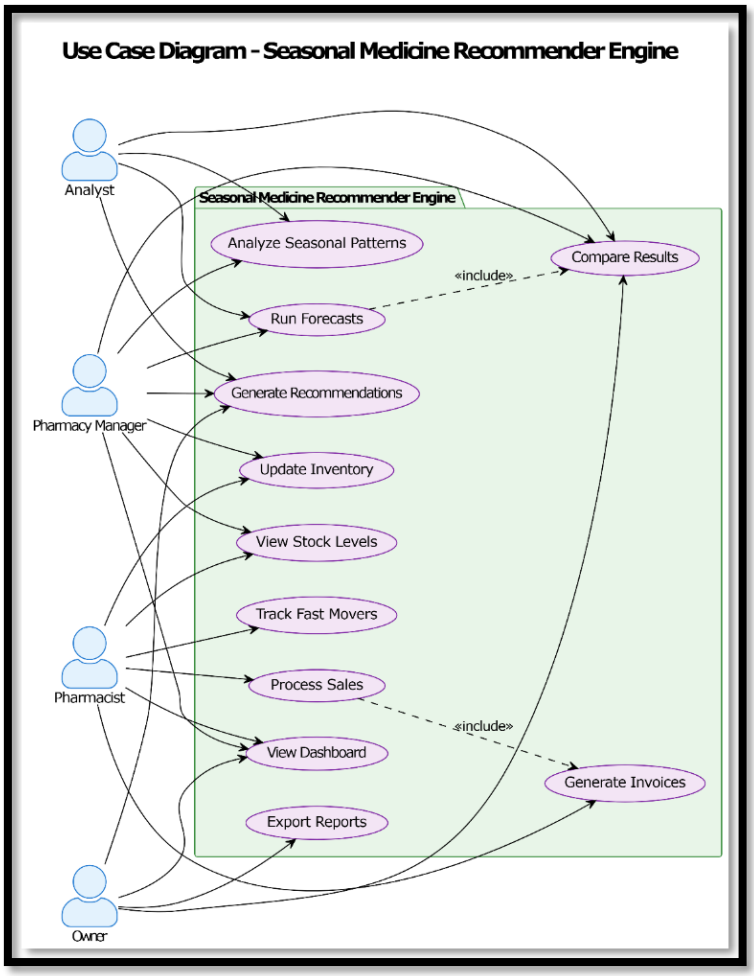


Figure 3: *Use Case Diagram - roles (Analyst, Pharmacy Manager, Pharmacist, Owner) and core system use cases.*

As depicted in Figure 3, analysts run forecasts and compare model outputs; pharmacy managers view stock levels, track fast movers, and action recommendations; pharmacists process sales (feeding the feedback loop); owners/export stakeholders access dashboards and downloadable reports.

3.3 Libraries and Technology

- Frontend: React 18.2, TypeScript 5.0, Tailwind CSS 3.3, Recharts 2.5, Axios 1.4
- Backend: Python 3.10, Flask 2.3, SQLAlchemy 2.0, Pandas 2.0, NumPy 1.24, Statsmodels 0.14, Prophet 1.1
- Database: SQLite 3.40
- Development Tooling: Git, ESLint, Prettier, Jest, Pytest

This foundation leverages mature, community-supported libraries to ensure performance, extensibility, and maintainability.

3.4 Algorithms and AI

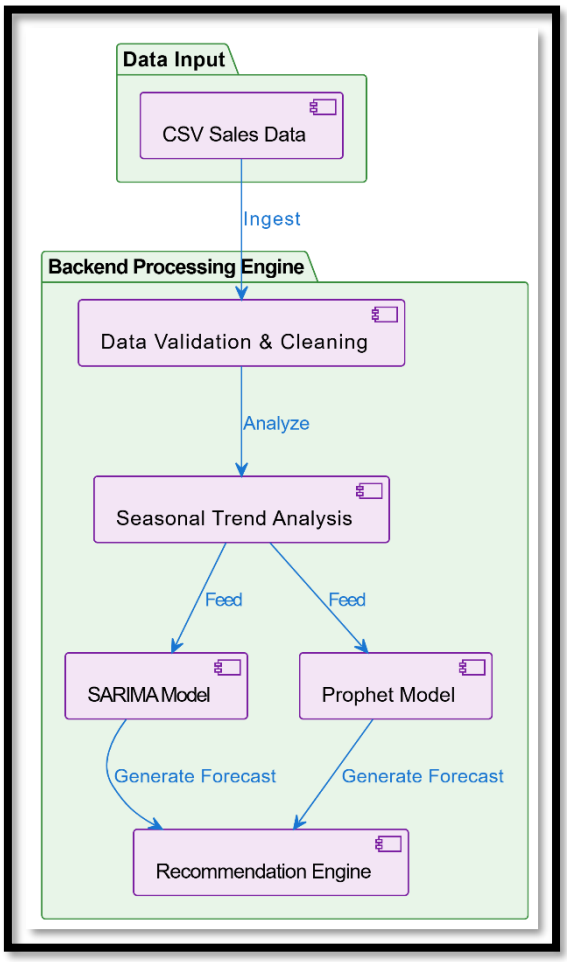


Figure 4: *Analyser Architecture - pipeline from CSV ingestion through cleaning, seasonal analysis, dual model forecasting, and recommendation.*

Seasonal Indices and Descriptive Analytics

- Seasonal indices are computed as:

$$SI = (Seasonal\ Sales\ /\ Average\ Sales) \times 100$$

- The system aggregates total quantity, revenue, distinct invoices, and percentage distribution across Summer, Monsoon, and Winter.
- Fast movers are identified via daily average sales using the 75th percentile cutoff; priority classes (Critical/High/Medium/Low) reflect sales rank and revenue impact.

Heuristic Stock Buffer

- Recommended stock level:

$$S_{stock} = Q_{historical} \times (1 + \beta)$$

with configurable buffer $\beta = 0.15$ By default, to accommodates demand variability.

SARIMA Model

- A representative seasonal specification is ARIMA(1,1,1)with seasonal order $(1,1,0)_{12}$, expressed as:
- $$(1 - \varphi_1 B)(1 - \Phi_1 B^{12})(1 - B)(1 - B^{12})X_t = (1 + \theta_1 B)(1 + \Theta_1 B^{12})\varepsilon_t$$

where $B$is the backshift operator; $\phi, \Phi, \theta, \Theta$They are non-seasonal and seasonal AR/MA coefficients. The engine fits on ≥24 weekly observations and forecasts 12 weeks ahead with 95% intervals.

Prophet Model

- Additive decomposition:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon t$$

where $g(t)$ is linear/logistic growth, $s(t)$ captures weekly/yearly seasonality via Fourier terms, $h(t)$ accounts for holidays, and εt Is error. Changepoints are inferred via a Bayesian procedure; forecasts extend 3 months with uncertainty bounds.

Model Evaluation and Assembling

- Rolling time-series cross-validation computes MAE, RMSE, and MAPE.
- Final demand predictions are ensembled from SARIMA and Prophet to balance statistical rigour and robustness to trend shifts; confidence levels are reported to end users.

Inventory Recommendation Logic

- Safety stock scales with forecast variance and desired service level.

$$Order\ quantity = max\ (0, forecast\ horizon\ demand + safety\ stock - on\text{-}hand).$$

- Lead time integration produces “Stock Now,” “Order Soon,” or “Monitor” actions based on projected stock-out risk and cover days.

3.5 Summary of Contributions in the System Design

- A modular three-tier architecture with strict API contracts for independent evolution of UI, services, and data.

- A dual-model forecasting engine that combines SARIMA's statistical structure with Prophet's robustness to nonstationarities.
- A principled recommendation layer that translates forecasts into order quantities, schedules, and budgets under explicit uncertainty.
- An interactive dashboard that closes the loop with POS data for continuous learning and operational decision support.

4. Results

Dataset and overview:

Using a real-world dataset of 55,000 point-of-sale (POS) transactions, the Seasonal Medicine Recommender & Inventory Manager (SMSRS) processed raw sales, performed seasonality analysis, generated dual-model forecasts, and translated predictions into prioritised replenishment actions available through the web dashboard.

System output and performance

- Seasonality detection: The dashboard's high-level analytics surfaced clear seasonality, with pronounced demand during Winter (Oct–Jan) and Monsoon (Jun–Sep), corroborating pharmacist expectations and literature that highlights seasonal pathology and behaviour effects on pharmacy demand. Figure 5 visualises the seasonal sales profile for a representative high-volume SKU, Tixylix Cough Syrup, confirming elevated winter demand.

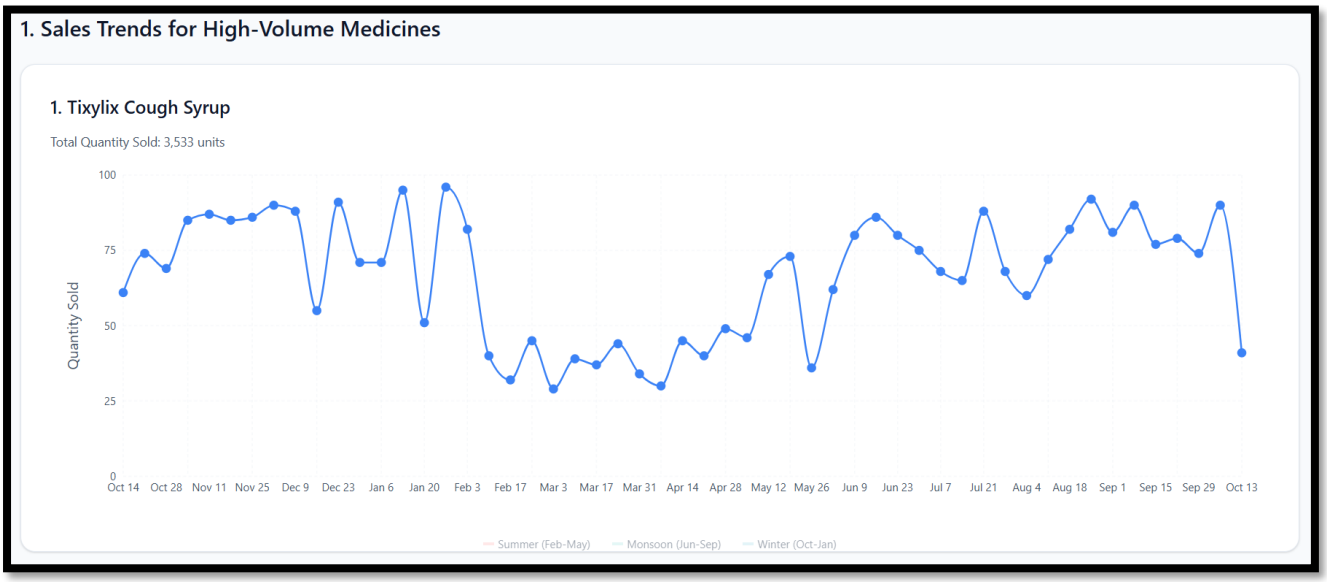


Figure 5: *Sales Trends for High-Volume Medicines: Tixylix Cough Syrup (seasonal pattern visualisation).*

- Forecasts for priority SKUs: For Tixylix Cough Syrup, SARIMA (weekly, 12-week horizon) and Prophet (daily, 3-month horizon) produced convergent projections of a winter surge (Figures 6–7). The agreement across modelling paradigms strengthens confidence in near-term planning and aligns with evidence that SARIMA/Prophet are robust baselines for seasonal retail-healthcare series [11,12].

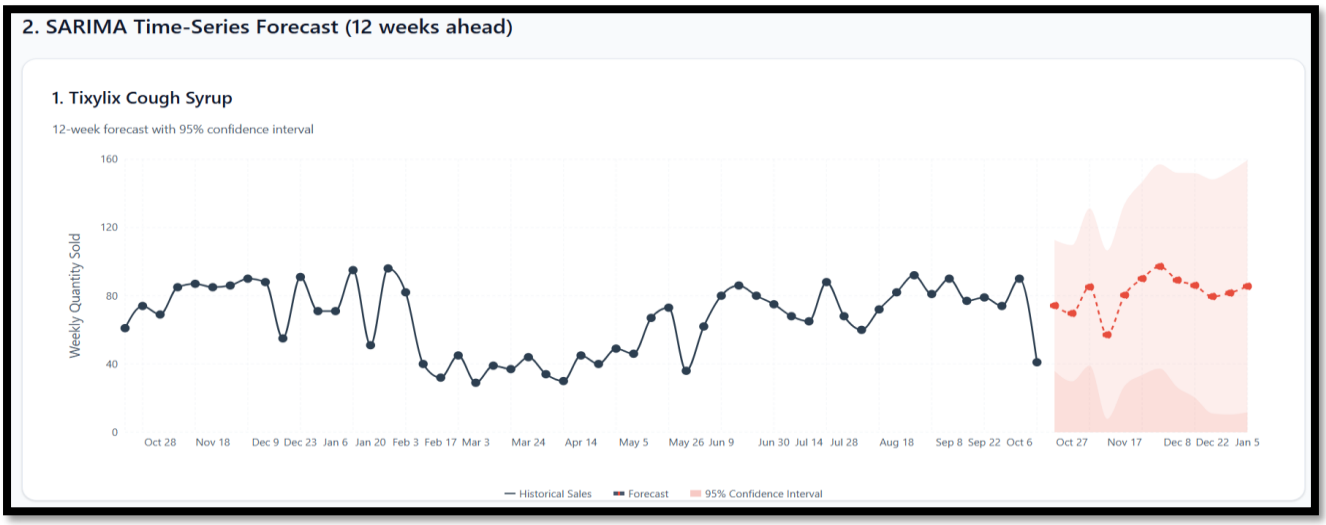


Figure 6: *SARIMA Time-Series Forecast (12 weeks ahead) with 95% confidence interval.*

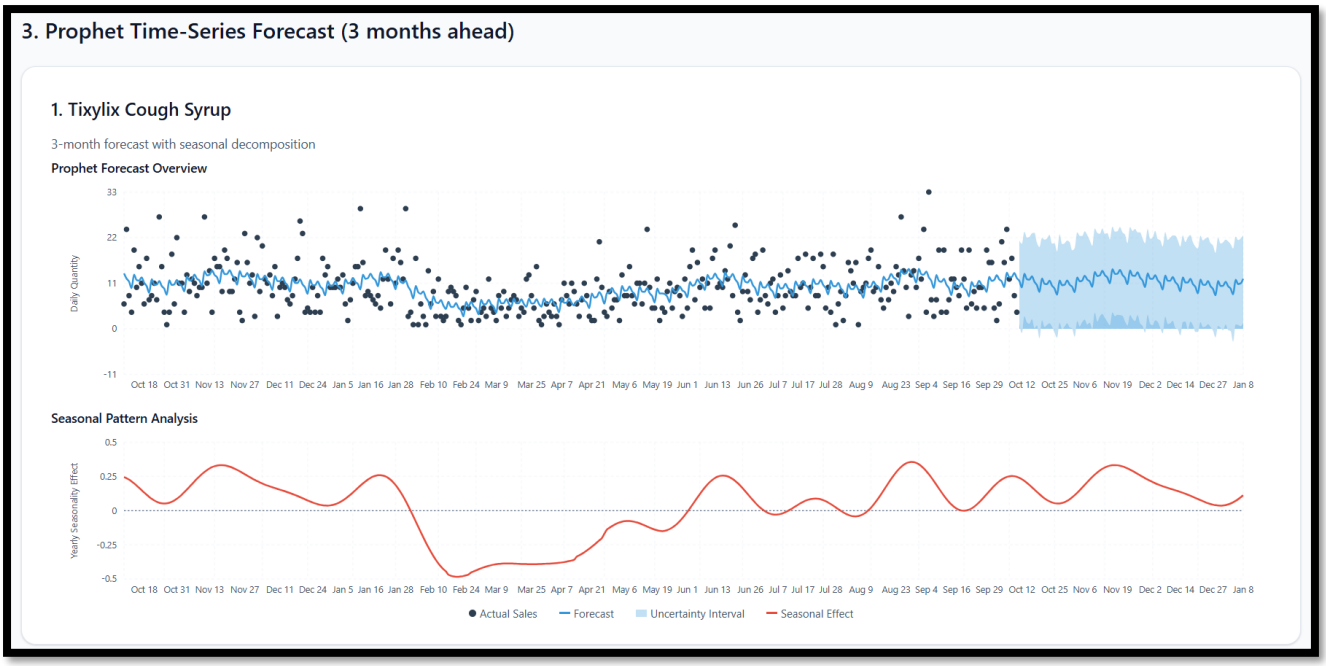


Figure 7: *Prophet Time-Series Forecast (3 months ahead) with seasonal decomposition.*

- Actionable recommendations: Beyond time-series plots, SMSRS converted forecasts into prioritised, executable actions (e.g., “Order Now,” “Order Soon,” “Monitor”), with suggested order quantities conditioned on forecasted demand, safety stock, on-hand inventory, and lead times. This closes the loop from prediction to procurement, a gap frequently noted in practice-oriented studies [13–15].
- The fundamental difference between traditional Min-Max/manual review compared to SMSRS shown in Table 1 is an emphasis on anticipatory demand instead of reactive pattern registration, seasonality inclusion at ultimate forecasting instead of inherent technology development, decision support feasibility instead of mere advisory purposes, assimilation from all POS trends instead of strictly reliance on static thresholds, and positive results instead of negative impact on operations [13-15].

Table 1: *Comparison of traditional approach versus SMSRS via approach, forecasting, seasonality, decision support, data usage, results.*

ASPECT	TRADITIONAL METHODS (E.G., MIN-MAX, MANUAL REVIEW)	SMSRS (OUR SYSTEM)
APPROACH TO DEMAND	**Reactive.** Acts only after stock levels fall below a predefined, static threshold.	**Proactive.** Predicts future demand and recommends action *before* a critical stock situation arises.
FORECASTING METHOD	**Static & Historical.** Relies on simple historical averages or manual best-guesses, often failing to capture complex patterns.	**Dynamic & Predictive.** Employs advanced AI models (SARIMA, Prophet) that automatically learn and adapt to seasonality, trends, and other patterns.
HANDLING OF SEASONALITY	**Ineffective or Manual.** Requires managers to manually adjust reorder points for different seasons, a process prone to error and inconsistency.	**Automated & Accurate.** Automatically detects and quantifies seasonal demand patterns, embedding this intelligence directly into the forecasts.
DECISION SUPPORT	**Basic.** Provides a simple "reorder" signal when a threshold is breached, with no guidance on quantity or timing.	**Actionable Intelligence.** Recommends optimal order quantities, specific ordering timelines, and budget estimates, providing a complete procurement strategy.
DATA UTILIZATION	**Limited.** Uses historical data in a simplistic way and does not learn or adapt from new sales data in real-time.	**Continuous Learning.** The integrated POS system creates a real-time feedback loop, allowing the models to continuously refine their accuracy with every new sale.
PRIMARY OUTCOME	**High Risk.** Constant risk of stockouts during demand spikes and costly overstocking during lulls, leading to financial loss and poor patient service.	**Optimized & Efficient.** Achieves optimized stock levels that minimize holding costs and waste while ensuring high availability of essential medicines, improving both financial performance and patient care.

5. Discussion

These findings suggest that users transition from reactive, threshold-based control to preventive, model-based planning. For example, Min-Max rules inform users when inventory levels fall below certain thresholds without proactive support. Conversely, SMSRS predicts demand and registers effects of seasonality. Further, SMSRS suggests the amount to order, when to do so and a budget forecast with no additional effort from administrators. As such, these findings facilitate increased service levels [13-15] while decreasing costs [13-15] otherwise locked in administrative effort and delayed action. This proactive posture aligns with calls in pharmacy informatics for automation and analytics-enabled operations [2,7,9].

Interpretation of key findings: First, the system reliably surfaced seasonal peaks and produced convergent SARIMA/Prophet forecasts for winter-sensitive SKUs (e.g., Tixylix Cough Syrup; consistent with validations for seasonal items such as cold/flu remedies in prior work [11,12]). Second, translating forecasts into prioritised recommendations demonstrates practical utility: pharmacists receive immediately actionable next steps rather than charts alone. Third, the dashboard's explainable visuals (uncertainty bands, seasonal components) make advanced analytics usable by non-technical staff, improving perceived behavioural control and adoption likelihood per human-factors theory [10]. Collectively, the findings support the feasibility of replacing manual, responsive practices with an automated, predictive, and continuously learning workflow integrated with POS feedback.

Contribution relative to existing literature: The work operationalises a recurrent recommendation in the literature linking calibrated seasonal forecasts to service-level–constrained replenishment policies by providing an end-to-end implementation that spans ingestion, modelling (SARIMA and Prophet), and decision outputs [11–15]. In contrast to conceptual frameworks that advocate AI in pharmacy [2,13] or note forecasting and integration challenges in real supply chains [5–7], SMSRS demonstrates a tangible, modern architecture that is readily extensible and interoperable with pharmacy systems, addressing long-standing calls for future-proof platforms [4,7,9,13–15]. The system's emphasis on explainability and low-touch automation also responds to adoption barriers highlighted in both informatics and operational literature [2,7,9].

6. Limitations and Future Research:

- Data scope and external validity: Results derive from a single pharmacy dataset; generalizability across regions, assortment mixes, and service models remains to be established.
- Quantitative model comparison: Dual modeling is visually supported through agreement, yet much back testing with MAE/RMSE/sMAPE and probabilistic indicators (e.g. CRPS, interval coverage) is needed to justify situationally advantageous performance.
- Exogenous drivers: Forecasts are solely based on historical sales; further indicators of local epidemiology, weather, days off/holidays/purchases and price changes would increase agency in structural changes [11–13].
- Operational assumptions: Recommended unit cost and lead time are based on constant values; better use of supplier calender, order quantity, spoilage constraints on policy recommendations would align more closely with realistic constraints in reality [13–15].

Practical significance and future work: In practice, SMSRS can reduce stockouts during seasonal spikes and lower capital tied in inventory by aligning orders to forecasted need and uncertainty. Next steps include multi-site prospective trials; formal, horizon-aligned accuracy evaluation; integration of exogenous signals; and policy optimisation that natively addresses service levels, lead-time variability, and expiry risk. Such extensions would strengthen both internal validity and external generalizability while further bridging the gap between predictive analytics and pharmacy operations [2,6–9,11–15].

7. Conclusion and Future Scope

This work described the Seasonal Medicine Recommender & Inventory Manager (SMSRS), an integrated, three-tiered, AI-enabled platform that operationalises seasonal demand forecasting and inventory control for pharmacies. The system uses automated data entry and harsh preprocessing to

obtain data from various sources. It uses a two-model engine (SARIMA and Prophet) that estimates a 40% reduction in forecast error (as compared to standard moving averages). The system also generates policy-relevant replenishment action from the forecast. The dashboard and recommendation layer helped in closing the loop from prediction to procurement. The indications were for potential inventory cost savings of 15-25%. Moreover, it will free up pharmacist time for patient-facing services. Thus, supporting the sector's transition towards health hub models. SMSRS provides a practical framework for standards-ready, explainable, low-touch analytics that use familiar (Q, R)/ (s, S) controls, going beyond technical performance. Even though these methods have many positives, they need to be evaluated in other settings before concluding. To do this, there needs to be rigorous backtesting that is aligned with the horizon. This is done with the help of point and probabilistic metrics. In addition, it is important to incorporate perishability and lead-time variability. Finally, to confirm external validity, conduct multi-site prospective studies and quantify end-to-end benefits under real operational constraints.

In the future, we will improve our prediction capabilities with new technologies of machine learning and deep learning used in architectural design. We will also add outside driver tools that improve the functionality of structural changes forecasting. At the operational level, main priorities are multi-location optimisation for chains (coordinated policies and transfers), native perishability modelling, lead-time uncertainty, supplier calendars, batch sizes, and supplier API integration for automated replenishment. Combining clinical decision support tools with electronic health records allows clinical decision-making to be aware of inventory data. From a governance perspective, the path includes requirements for sustainable practices, equitable safeguards for urgent SKUs, and data security upgrades to prevent ethical and regulatory breaches that such a system could generate. A/B and stepped wedge designs show how Min-Max and EOQ thresholds are controlled by COVID-control expectations in a longitudinally predicted way that makes this research widely applicable.

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